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Water sprinkler

Abstract

The present application relates to a water sprinkler, including a housing, a solar module, a battery module, a water inlet/outlet pipe, and a control module, where the water inlet/outlet pipe and the control module are arranged in the housing, the solar module is arranged outside the housing and rotatably connected to the housing, the water inlet/outlet pipe is provided with at least one control valve, the control module is in communication connection with the control valve and controls the opening and closing of the control valve, and the solar module, the control module and the control valve are each electrically connected to the battery module.

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Background/Summary

TECHNICAL FIELD

(1) The present application relates to the technical field of water sprinklers, and in particular to a water sprinkler.

BACKGROUND

(2) Generally, watering is manually controlled. However, with urban greening and continuous enlargement of horticultural scale, traditional manual watering methods cannot meet the requirements for efficient and precision irrigation.

(3) Therefore, there are proposed a number of automatic water sprinklers on the market.

SUMMARY

(4) The present application provides a water sprinkler, including a housing, a solar module, a battery module, a water inlet/outlet pipe, and a control module, where the water inlet/outlet pipe and the control module are arranged in the housing, and the solar module is arranged outside the housing and rotatably connected to the housing; the water inlet/outlet pipe is provided with at least one control valve, and the control module is in communication connection with the control valve and controls the opening and closing of the control valve; the solar module, the control module and the control valve are each electrically connected to the battery module; and the water inlet/outlet pipe is provided with a water inlet and at least one water outlet, the number of the control valves matches the number of the water outlets, and the control valve controls the opening and closing of a corresponding water outlet; the water sprinkler further comprising a charging interface module electrically connected to the battery module; where the housing includes a front housing and a rear housing, the front housing being detachably connected to the rear housing, the front housing being provided with a mounting groove, and the solar module and the charging interface module being both arranged in the mounting groove; and the solar module includes a solar panel, a mounting base and a rotating member, the solar panel being mounted on the mounting base and electrically connected to the battery module, and the mounting base being rotatably connected to the front housing by the means of the rotating member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to more clearly describe the technical solutions in the embodiments of the present application or the prior art, the drawings necessary for describing the embodiments or the prior art will be briefly described below. Apparently, the drawings in the description below merely show some of the embodiments of the present application, and those of ordinary skill in the art would have obtained other drawings from these drawings without involving any inventive effort.

(2) FIG. 1 is a first structural schematic perspective view of a water sprinkler according to an embodiment of the present application.

- (3) FIG. 2 is a second structural schematic perspective view of an water sprinkler according to the embodiment of the present application.
- (4) FIG. 3 is a schematic diagram of a functional module of the water sprinkler according to an embodiment of the present application.
- (5) FIG. 4 is an exploded structural schematic view of the water sprinkler according to an embodiment of the present application.
- (6) FIG. 5 is a partial structural schematic view of the water sprinkler according to an embodiment of the present application.
- (7) FIG. 6 is a structural schematic perspective view of a water inlet/outlet pipe of a water sprinkler according to an embodiment of the present application.
- (8) FIG. 7 is a structural schematic sectional view of a water inlet/outlet pipe of a water sprinkler according to an embodiment of the present application.

LIST OF REFERENCE SIGNS

- (9) **1.** Water sprinkler; **11.** Housing; **12.** Solar module; **13.** Battery module; **14.** Charging interface module; **15.** Water inlet/outlet pipe; **16.** Control module; **17.** Flow sensor; **18.** Soil moisture sensor; **19.** Weather sensor; **111.** Front housing; **112.** Rear housing; **113.** Physical key; **151.** Control valve; **152.** Water inlet; **153.** Water outlet; **121.** Solar panel; **122.** Mounting base; **123.** Rotating member; **161.** Wireless communication module; **1111.** Mounting groove.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) For ease of understanding of the present application, a more complete description of the present application will be made with reference to relevant drawings. Embodiments of the present application are illustrated in the accommodating drawings. However, the present application may be implemented in many different forms and is not limited to the embodiments described herein. On the contrary, these embodiments are intended to make the disclosure of the present application more thorough and comprehensive.

(11) Unless otherwise defined, all technical and scientific terms used herein shall have the same meanings as commonly understood by those skilled in the art to which this application relates. The terms used herein in the specification of the present application are only used to describe specific embodiments, and are not intended to limit the present application.

(12) It should be understood that the singular forms of “a”, “an” and “the/this” used in the present application may also include plural forms, unless clearly indicated otherwise in the context. It should also be understood that the terms such as “include/comprise” and “have” specify the presence of mentioned features, units, steps, operations, components, portions, or combinations thereof, but do not exclude the presence or addition of one or more other features, units, steps, operations, components, portions, or combinations thereof.

(13) It should be noted that when an element is referred to as being “fixed to” another element, the element may be directly located on the other element or an intermediate element may exist. When one element is considered to be “connected” to another element, the element may be directly connected to another element or an intermediate element may also be present. The terms “vertical”, “horizontal”, “left”, “right” and similar expressions used herein are for the purpose of illustration only.

(14) In the present application, orientations or position relationships indicated by terms such as “upper”, “lower”, “left”, “right”, “front”, “rear”, “top”, “bottom”, “inner”, “outer”, “middle”, “vertical”, “horizontal”, “transverse” and “longitudinal” are based on orientations or position relationships shown in the accommodating drawings. These terms are mainly intended to better describe the present application and the embodiments thereof and are not intended to limit that a mentioned device, element or constituent part must have a specific orientation, or be constructed and operated in a specific orientation. Moreover, some of the above terms may be used to denote other meanings apart from the orientations or positional relationships. For example, the term “upper” may also be used to denote a certain attachment relationship or connection relationship in

some circumstances. For those skilled in the art, the specific meanings of the terms mentioned above in the present application may be construed according to specific circumstances.

(15) In addition, the terms “mount”, “arrange”, “provide”, “connect” and “couple” should be understood in a broad sense. For example, it may be a fixed connection, a detachable connection, or an integral construction, may be a mechanical connection, or an electrical connection, or may be a direct connection, an indirect connection by means of an intermediate medium, or internal communication between two means, elements or constituent parts. The specific meanings of the above terms in the present application may be understood by those skilled in the art according to specific situations.

(16) Referring to FIGS. 1-4, the embodiments of the present application provide a water sprinkler **1**. The water sprinkler **1** includes a housing **11**, a solar module **12**, a battery module **13**, a charging interface module **14**, a water inlet/outlet pipe **15** and a control module **16**.

(17) The water inlet/outlet pipe **15** and the control module **16** are both arranged in the housing **11**, the charging interface module **14** is exposed from the housing **11**, and the solar module **12** is arranged outside the housing **11** and rotatably connected to the housing **11**. The water inlet/outlet pipe **15** is provided with at least one control valve **151**, and the control module **16** is in communication connection with the control valve **151** and controls the opening and closing of the control valve **151**. The charging interface module **14**, the solar module **12**, the control module **16** and the control valve **151** are each electrically connected to the battery module **13**.

(18) Optionally, the charging interface module **14** is a Type-C interface module, a Micro USB interface module, or a Lighting interface module.

(19) Specifically, the battery module **13** is a supercapacitor or an ordinary rechargeable battery.

(20) Specifically, the control valve **151** is an electric valve.

(21) In some embodiments, the housing **11** is further provided with physical keys **113** connected to the control valves **151**, and the control valves **151** are operated by the physical keys **113** to open and close.

(22) It should be understood that the control module **16**, as a core of the whole system of the water sprinkler **1**, is responsible for receiving signals, executing control algorithms and sending control instructions.

(23) It should be understood that the solar module **12** uses a high-efficiency solar panel to convert solar energy into electrical energy, which is stored in the battery module **13** to ensure the continuous operation of the water sprinkler **1** during the day and when sunlight is sufficient. The charging interface module **14** performs fast charging with a standard power adapter, is suitable for use in situations where solar energy is insufficient, for example, at night or in cloudy days, to ensure stable power supply for the water sprinkler **1**.

(24) The water sprinkler **1** according to the embodiments of the present application is powered with solar energy, the problem is solved that an existing watering system relies on a fixed power source in most cases, is high in mounting and maintenance costs, is difficult to operate for a long period of time in an environment where energy supply is unstable, and thus causes poor irrigation effect and low irrigation efficiency, stable power support can be provided for the water sprinkler **1**, long-term stable operation of the water sprinkler **1** in various environments is ensured, and thus the irrigation effect and irrigation efficiency of the water sprinkler **1** are improved. In addition, the opening and closing of the control valve **151** are controlled by the control module **16**, and the opening and closing of the water inlet/outlet pipe **15** can thus be controlled, so that the intelligentization of the water sprinkler **1** can be improved, and intelligent irrigation according to actual requirements is facilitated. Moreover, the provision of the charging interface module **14** enables dual-mode power supply based on solar energy and an external power supply can be achieved, and the long-term stable operation of the water sprinkler **1** in various environments is further ensured.

(25) As shown in FIGS. 4 and 5, the housing **11** includes a front housing **111** and a rear housing **112**, and the front housing **111** is detachably connected to the rear housing **112**. The front housing

111 is provided with a mounting groove **1111**, and the solar module **12** and the charging interface module **14** are both provided in the mounting groove **1111**.

(26) Specifically, the mounting groove **1111** matches the solar module **12** in shape and size to facilitate the storage of the solar module **12** in the mounting groove **1111**.

(27) In the embodiments of the present application, the assembly and later maintenance of the water sprinkler **1** are facilitated by providing the front housing **111** and the rear housing **112** that are detachably connected. Furthermore, the solar module **12** and the charging interface module **14** are assembled by providing the mounting groove **1111**, so that the integration level of the water sprinkler **1** can be improved, reducing the overall size of the water sprinkler **1**.

(28) Specifically, the solar module **12** includes a solar panel **121**, a mounting base **122** and a rotating member **123**, where the solar panel **121** is mounted on the mounting base **122** and electrically connected to the battery module **13**, and the mounting base **122** is rotatably connected to the front housing **111** by means of the rotating member **123**.

(29) In some embodiments, the rotating member **123** is a hinge **1232**, and a connection is made by means of the hinge **1232** to make the assembly simple, convenient and quick, so that an angle of rotation of the solar panel **121** can be achieved by manually turning the mounting base **122**.

(30) In some embodiments, the rotating member **123** is an electric motor **1231**, the electric motor **1231** is in communication connection with the control module **16**, and the control module **16** controls the angle of rotation of the solar panel **121** by means of the electric motor **1231**, so that the water sprinkler is more intelligent and automated.

(31) As shown in FIGS. **6** and **7**, the water inlet/outlet pipe **15** is provided with a water inlet **152** and at least one water outlet **153**, where the water inlet **152** is arranged on one side of the water inlet/outlet pipe **15**, and the water outlet **153** is arranged on the opposite side of the water inlet/outlet pipe **15**. The number of the control valves **151** matches the number of the water outlets **153**, and the control valves **151** control the opening and closing of corresponding water outlets **153**.

(32) Specifically, the water inlet/outlet pipe **15** is provided with four water outlets **153**, the water inlet/outlet pipe **15** is provided with four control valves **151**, and the four control valves **151** are independent of each other and respectively control the opening and closing of the four water outlets **153**.

(33) Specifically, the four water outlets **153** are arranged in parallel.

(34) In some embodiments, a spray head is assembled at each of the water outlets **153** to support the switching of different types of spray heads and meet diversified horticultural requirements.

(35) In some embodiments, the four control valves **151** independent of each other are provided to control the four water outlets **153** to allow the four water outlets **153** to be controlled respectively according to control instructions. According to the requirements of different plants or regions, the volume and time of the irrigation are set respectively to achieve precision irrigation, such that four-channel intelligent watering of the water sprinkler **1** is thus achieved, the irrigation efficiency is improved, water resources are also saved, and operation procedures are simplified.

(36) Still referring to FIGS. **3**, **4**, **6** and **7**, the water sprinkler **1** further includes flow sensors **17**, where the number of the flow sensors **17** matches the number of the water outlets **153**, the flow sensors **17** are in communication connection with the control module **16**, and the flow sensors **17** are configured to monitor the water flows of corresponding water outlets **153**. With this configuration, a corresponding watering volume can be set according to actual requirements, it can be ensured that the irrigation volume meet the set requirements, and the irrigation efficiency can be further improved.

(37) It can be understood that a plurality of flow sensors **17** achieve the monitoring of a plurality of water outlets **153**, and each of the flow sensors **17** monitors a corresponding one of the water outlets **153**.

(38) Specifically, in some embodiments, the number of the flow sensors **17** is four, the number of the water outlets **153** is four, and the four flow sensors **17** similarly control the opening and closing

of the corresponding water outlets **153**.

(39) In addition, the flow sensors **17** can also detect abnormalities such as water leakage or blockage, and alert a user with a system alarm.

(40) The water sprinkler **1** further includes a soil moisture sensor **18** in communication connection with the control module **16** and configured to monitor soil moisture. With such a configuration, an environment can be monitored in real time to intelligently adjust irrigation plans and save water resources.

(41) The water sprinkler **1** further includes a weather sensor **19** in communication connection with the control module **16** and configured to monitor ambient temperature, ambient humidity and ambient rainfall. With such a configuration, the environment can be monitored in real time to avoid irrigation during rainfall, further saving the water resources.

(42) It should be understood that the battery module **13** is electrically connected to the control module **16**, the flow sensors **17**, the soil humidity sensor **18** and the weather sensor **19** to supply power for the entire system of the water sprinkler **1**.

(43) The control module **16** is integrated with a wireless communication module **161**, and the wireless communication module **161** is at least one of a WiFi module, a Bluetooth module and a Zigbee module. By being integrated with the wireless communication module **161**, the water sprinkler **1** can be wirelessly connected to a smart phone, a tablet computer, etc., such that the water sprinkler **1** can perform two-way data interaction with a mobile device of the user to achieve remote management and monitoring.

(44) The embodiments of the present application further provide an operation interface for the user by means of a liquid crystal display screen or a mobile application, allowing the user to set the irrigation plans for channels and view real-time data and system status.

(45) The technical features of the above embodiments may be combined arbitrarily. For the purpose of brevity of description, all the possible combinations of the technical features in the above embodiments are not described. However, as long as there is no contradiction between the combinations of these technical features, they shall all fall within the scope of the description.

(46) The embodiments described above merely illustrate several implementations of the present application and are described relatively specifically and in detail, but should not be construed as limiting the patent scope of the present application. It should be noted that several variations and improvements may also be made by those of ordinary skill in the art without departing from the concept of the present application, and should fall within the scope of protection of the present application. Therefore, the scope of protection of the present application shall be defined by the appended claims.

Claims

1. A water sprinkler, comprising a housing, a solar module, a battery module, a water inlet/outlet pipe, and a control module, wherein the water inlet/outlet pipe and the control module are arranged in the housing, and the solar module is arranged outside the housing and rotatably connected to the housing; the water inlet/outlet pipe is provided with at least one control valve, and the control module is in communication connection with the control valve and controls the opening and closing of the control valve; and the solar module, the control module and the control valve are each electrically connected to the battery module; and the water inlet/outlet pipe is provided with a water inlet and at least one water outlet, the number of the control valves matches the number of the water outlets, and the control valve controls the opening and closing of a corresponding water outlet; the water sprinkler further comprising a charging interface module electrically connected to the battery module; wherein the housing comprises a front housing and a rear housing, the front housing being detachably connected to the rear housing, the front housing being provided with a mounting groove, and the solar module and the charging interface module being both arranged in

the mounting groove; and the solar module comprises a solar panel, a mounting base and a rotating member, the solar panel being mounted on the mounting base and electrically connected to the battery module, and the mounting base being rotatably connected to the front housing by the means of the rotating member.

2. The water sprinkler of claim 1, wherein the rotating member is a hinge.

3. The water sprinkler of claim 1, wherein the rotating member is an electric motor in communication connection with the control module, and the control module controls an angle of rotation of the solar panel by means of the electric motor.

4. The water sprinkler of claim 1, further comprising flow sensors, the number of the flow sensors matching the number of the water outlets; and the flow sensors being in communication connection with the control module, and configured to monitor water flows of corresponding water outlets.

5. The water sprinkler of claim 1, further comprising a soil moisture sensor, the soil moisture sensor being in communication connection with the control module and configured to monitor soil moisture.

6. The water sprinkler of claim 1, further comprising a weather sensor, the weather sensor being in communication connection with the control module and configured to monitor ambient temperature, ambient humidity and ambient rainfall.

7. The water sprinkler of claim 1, wherein the control module is integrated with a wireless communication module, the wireless communication module being at least one of a WiFi module, a Bluetooth module and a Zigbee module.
